

Property Rights Restrictions and Self-Employed Workers: Evidence From For-Hire Owner–Operators in U.S. Trucking

Martin Holz hacker¹ , Harlow Loch², Jason Miller¹ and Alex Scott³

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Abstract

In 2019, self-employed individuals in the United States owned and operated more than 27 million nonemployer firms and generated more than \$1.3 trillion in revenue. Our study examines how self-employed workers adjust their operations in response to an exogenous regulatory change that places limits on how they can utilize their productive assets. Extending modern property rights theory of the firm (PRTF), we studied how self-employed for-hire truck drivers modified their operations in response to the required installation of electronic logging devices (ELDs) in their vehicles. The introduction of ELDs bolstered enforcement of preexisting legal hours-of-service limits, thereby weakening the drivers' ability to accept hauls above statutory hours-of-service limits and reducing the strength of their economic property rights. Our difference-in-differences analysis that relies on self-employed truck drivers exempt from the ELD regulation as a control group reveals that the restrictions imposed by the ELD mandate, while effective at curbing major hours-of-service violations, prompted a change in a host of market behaviors among affected drivers, including a shift toward working as subcontractors, decreased labor supply, and increased exit rates from the industry. These findings contribute to the literature on PRTF. First, our findings underscore the significance of economic property rights in shaping self-employed workers' operating decisions as legal property rights were largely left unchanged under the ELD regulation. Second, our findings with respect to subcontracting highlight that property rights restrictions placed on asset attributes may prompt self-employed workers to unbundle economic property rights and allocate them to exchange partners better equipped to maximize the economic value reaped from these rights. Such transfer of economic property rights common to intermediate forms of organizational design such as subcontracting embodies a multidimensional view of economic property rights. By demonstrating a range of consequences of restricting self-employed workers' economic property rights, our study carries important implications for theory, managers, and policymakers.

Keywords

Property Rights, Misconduct, Self-Employment, Subcontracting, Organizational Design

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1 Introduction

Self-employed workers running nonemployer firms—firms with no one on their payroll—play a critical role in many industry sectors in the United States (Davis et al., 2009). In 2019, the Census Bureau (2019) reported 27.1 million nonemployer firms, whose aggregate revenue surpassed \$1.3 trillion. These firm number and revenue estimates are up 24.9% and 45.3% from 2009 estimates, respectively (Census Bureau, 2009). The sharp rise in nonemployer firms aligns with employer firms' increasing reliance on contract labor (e.g., Katz and Krueger, 2019; Oyer, 2016) including free-lancers, software developers, influencers, rideshare drivers, marketers, and consultants (MBO Partners, 2021).

As independent business owners, these workers rely on their ability to utilize the tangible and intangible assets under their control. Asset ownership confers property rights that provide both control and incentives for self-employed workers (Grossman and Hart, 1986; Holmstrom and Milgrom,

¹Michigan State University, East Lansing, MI, USA

²Franklin & Marshall College, Lancaster, PA, USA

³University of Tennessee, Knoxville, TN, USA

Corresponding author:

Martin Holz hacker, Michigan State University, 632 Bogue St, N256, East Lansing, MI 48824, USA.

Email: mkh@msu.edu

1994). While prior research highlights the importance of protecting property rights through government enforcement (Johnson et al., 2002), little is known regarding how self-employed workers respond to rule and regulatory changes that restrict their property rights. Investigating this topic is of utmost importance both theoretically and practically. Theoretically, the convergence of asset owner and operator in self-employment settings bypasses complexities associated with the owners' delegation of decision rights for the use of assets to agents acting on their behalf (Jensen and Meckling, 1976). Since the delegation process in larger firms entails some loss of effective control of productive assets (Foss and Foss, 2015), examining self-employed workers is ideally suited for elaboration (Fisher and Aguinis, 2017) of the property rights theory of the firm (PRTF) (Grossman and Hart, 1986; Hart and Moore, 1990). Practically, the sharp increase in nonemployer firms and revenue coupled with the important share of activity conducted by self-employed workers in industries such as real estate, professional services, construction, and transportation make understanding the impact that restricting property rights can have on self-employed workers important.

This research takes an initial step in examining how self-employed workers respond to changes in the enforcement of government rules that restrict their property rights by curtailing how they can use their assets. We do this by modifying modern PRTF in two ways. As a first modification, we build on the distinction between legal property rights (LPRs) and economic property rights (EPRs). As Kim and Mahoney (2005) and Foss and Foss (2015) note, modern PRTF relies on a legal definition of property rights as it assumes that legal ownership of an asset is perfectly enforced and conveys control over an asset unless otherwise voluntarily surrendered to some other party involved in exchange (Grossman and Hart, 1986). We adopt the economic definition of property rights defined by Barzel and Allen (2023: 15)—writing outside the PRTF tradition—as an “individual’s ability (in expected terms) to exercise a choice with respect to a commodity or some other thing” that includes various uses, such as consumption, production, and transfer, among others.

The distinction between EPRs and LPRs we make is not just a semantical one due to EPRs' alleged greater impact on expectations underlying utility or income maximizing behavior (Barzel and Allen, 2023: 66–67). As a second modification, we replace PRTF's unidimensional view of asset ownership—which stems from its inherent emphasis on LPRs (Demsetz, 1998)—with a multidimensional view that emphasizes how restrictions that reduce the strength of EPRs regarding one type of choice (e.g., how many hours a day a truck can drive) influence the willingness of self-employed workers to unbundle and cede other EPRs to another party (e.g., shifting from self-employment to subcontracting with larger carriers). This change follows from Foss and Foss (2022: 58), emphasizing that “[economic] property rights... refer to an individuals' expected opportunity set.”

To test our theory, we analyzed a major regulatory change in the truck transportation industry (NAICS 484) between 2015 and 2018 when the Department of Transportation announced and implemented a policy requiring most commercial trucks to be equipped with electronic logging devices (ELDs) (Scott et al., 2021). The introduction of ELDs strengthened enforcement of preexisting federal hours-of-service (HOS) regulations that prohibit excessively long driving to reduce the risk of traffic accidents due to trucker fatigue. Before the ELD mandate, truck drivers were able to manipulate paper logs to exceed HOS limits undetected (Levy, 2023). The improved enforcement of HOS regulations under the ELD mandate diminished drivers' expectations of their own ability to accept and service long hauls. Thus, the mandatory introduction of ELDs is an ideal setting for our study as it provides a plausibly exogenous restriction to drivers' EPRs concerning the hours they work without material change to their LPRs that had been already curtailed by long-standing HOS regulations.

A carveout in the ELD regulation that exempted trucks with engines manufactured before the year 2000—necessary because engines manufactured in 1999 or earlier lack engine control modules (ECMs) needed for ELDs to communicate with a truck's engine (FMCSA, 2018, FMCSA 49 C.F.R. §395)—enabled us to conduct a difference-in-differences (DiD) analysis (Angrist and Pischke, 2009). Given the close overlap between a truck's model year (which is collected by roadside inspectors tasked with overseeing carrier safety) and its engine year, we can distinguish self-employed truck drivers—termed for-hire owner-operators (FHOOs)—affected by the ELD mandate from those who are not. The availability of a control group unaffected by the regulation guards our inferences against bias from aggregate time-varying market dynamics such as trends in demand or service prices that are potentially correlated with labor market outcomes for FHOOs (Angrist and Pischke, 2009: 171). Restricting our sample to older trucks within a narrow 3-year bandwidth of engine years around the mandate cutoff year further ensures that treatment and control group are comparable and subject to the same common trends, a key identifying assumption of our research design that we validate in additional tests.

Our DiD analysis revealed that the ELD mandate shaped FHOOs' behaviors along multiple dimensions. We found that treated FHOOs reduced major HOS violations, consistent with the stated purpose of the ELD mandate. Treated FHOOs also reduced their volume of labor supplied, which substantiates our argument that curtailing operating flexibility by reducing EPRs diminishes the incentive to produce. We also found that treated FHOOs show an elevated risk of ceasing operations permanently. Lastly, we found that treated FHOOs are more likely to change their organizational form by becoming subcontractors, especially for larger trucking firms.¹ This aligns with self-employed workers partitioning the EPR bundle and ceding the right to accept freight loads to larger carriers while retaining control over some of the other attributes of their asset (e.g., how to operate or maintain their truck). This finding

provides a direct test of the hypothesis at the heart of property rights economics that the distribution of property rights will “maximize the joint wealth net of transaction costs of the individuals involved” (Barzel and Allen, 2023: 67).

This work makes multiple contributions to both theory and practice. Distinguishing between EPRs and LPRs expands PRTF’s dynamic consilience (Thagard, 1978) by expanding the array of phenomena the theory can explain. The modern version of PRTF can only be used to examine how asset owners’ actions adjust to external shifts that affect legal rights over assets, which inherently limits applicability (Allen and Barzel, 2016; Demsetz, 1998; Foss and Foss, 2022). Relatedly, this helps address a critique of PRTF that its unidimensional conceptualization of property rights makes it inadequate to explain intermediate forms of organization that involve unbundling and partitioning of EPRs (Barzel and Allen, 2023; Foss and Foss, 2015; Kim and Mahoney, 2005). For example, within our framework, the self-employed worker’s decision to enter a subcontracting relationship involves voluntarily surrendering EPRs associated with independently soliciting business. Doing so allows the self-employed worker to take advantage of larger entities’ superior abilities to obtain business and leverage economies of scope that increase productivity and maximize the value of a specific EPR (Rakowski, 1988). Consistent with the description of EPRs as concerning an individual’s expected opportunity set in Foss and Foss (2022), we document how a regulatory change that reduces EPRs in one dimension can instigate a variety of behavioral changes by self-employed workers.

We also contribute to the literature on corporate misconduct (e.g., Campbell and Shang, 2022; Heese and Pérez-Cavazos, 2020) by offering a property rights angle. Understanding how management control systems and external scrutiny by regulators and media can curb misconduct has been the focus of several recent studies in accounting and finance. Our study takes a decidedly different approach by examining some of the adverse consequences of scrutinizing self-employed workers. By shedding light on some of the costs of imposing limits on self-employed workers’ EPRs, our findings have implications for both firms and regulators. For example, extension of these rules cause self-employed workers to exit the market, and social welfare may be negatively affected if buyers of self-employed workers’ goods or services, such as shippers buying trucking capacity (Scott, 2019; Scott et al., 2020), face labor shortages or must pay more for these services. Our results also elucidate how EPR restrictions can affect organizational design by pushing workers from self-employment to subcontracting. This shift embroils firms in a myriad of issues associated with worker misclassification, tax compliance, liability for torts committed by subcontractors, and concerns about the abuse of these workers (Masten, 1988; Weil, 2014). In conclusion, our study documents that EPR restrictions, while effective at deterring misconduct, may be socially costly in terms of fueling shortages of self-employed labor and spurring avoidance behaviors.

The remainder of this research is structured in five sections. Property Rights Theory of the Firm (PRTF) section reviews relevant property rights literatures. Institutional Background and Hypotheses section introduces the research setting and leverages PRTF to sketch the logic for the hypothesized effects. Data, Research Design, and Sample section explains the research design, data sources, and variables. Findings section presents the econometric specifications and reports results. Discussion section explains theoretical contributions, describes managerial and policy implications, notes limitations, and suggests future extensions.

2 Property Rights Theory of the Firm (PRTF)

2.1 Modern PRTF

At the heart of the modern PRTF (Grossman and Hart, 1986; Hart and Moore, 1990) is the notion that (legal) asset ownership confers a comprehensive set of property rights, most notably control over assets, and thereby also control over workers who operate them. For example, a motor carrier can hire a subcontractor who drives a truck that he owns or, alternatively, buy a truck and hire an employee to operate the truck per the carrier’s directions (Baker and Hubbard, 2004). A core premise of PRTF is that asset ownership generates incentives to undertake noncontractible investments because ownership conveys residual control rights (Grossman and Hart, 1986). PRTF predicts that the party whose noncontractible investments have a greater marginal impact on the joint surplus from exchange should own assets needed to generate products or services being exchanged (Woodruff 2002). Empirical evidence tends to be scant with a few notable exceptions from trucking (Baker and Hubbard 2003, 2004) and international trade (Feenstra and Hanson, 2005) that support the notion that the party concerned with ex-post expropriation under incomplete contracts holds the ownership of the assets in question.

Importantly for our study, modern PRTF grounds its notion of property rights in the legal realm only (Foss and Foss, 2022). Additionally, property rights under PRTF are monolithic in that ownership is assumed to confer an all-encompassing set of rights (Kim and Mahoney, 2005). Rooted in a review of pertinent property rights literature, we next introduce the notion of EPRs that challenges these viewpoints and enables us to derive several proposed modifications to modern PRTF.

2.2 Economic Property Rights (EPRs)

A long-standing literature study in economics has viewed property rights as *de facto* rights to an asset (Alchian, 1965; Cheung, 1970; Demsetz, 1967). Barzel and Allen (2023) define economic property rights (EPRs) as an actor’s perception that they can execute a choice regarding a commodity. Such choices “might include various uses

(consumption, destruction, improvements, possession, enjoyment, production), exchange (selling, buying, mortgaging, leasing), transfer (bequeathing, donating, purchasing), and importantly, the exclusion of others” (Barzel and Allen, 2003: 15–16). Whether actors expect they can exercise a choice depends on many factors, such as their efforts in protecting their assets; others’ attempts to appropriate their assets; governmental regulations; and enforcement through courts, police, or even the military. Per Barzel and Allen (2023: 28), we conceptualize the strength of an EPR as an actor’s perceptions regarding the expected probability they can carry out a choice with respect to an attribute of a resource. In most real-world settings, this probability is less than one, meaning EPRs are imperfect (Barzel and Allen, 2023: 27–29).

EPRs should be distinguished from legal property rights (LPRs) that merely grant an individual the “authority under the law to exercise a choice with respect to something” (Barzel and Allen, 2023: 18). While LPRs may enhance EPRs, the overlap between the two concepts is highly imperfect. One reason cited for the divergence of EPRs and LPRs is the lack of enforcement of legal property rights (Cheung, 1970). For example, the state may fail to enforce LPRs of retail store owners by not sufficiently protecting them from shoplifting. When this happens, the risk of theft depresses legal owners’ *de facto* expectations to remain in control of property, thereby weakening EPRs.

EPRs are best thought of as a bundle of various rights to an asset (Alchian and Demsetz, 1973; Demsetz, 1967). In line with this thinking, Barzel and Allen (2023: 25–27) conceptualize property rights for a resource as a matrix of property rights for its various uses and attributes. For a FHO, this may include various rights of usage, such as the ability to drive and maintain a truck, the rights to profits (i.e., residuals) generated from its various uses, the right to transfer legal ownership, and the ability to exclude others from using the truck. However, legal owners do not hold all EPRs over an asset. A carrier who hires an employee to drive a truck effectively delegates EPRs to use the company-owned truck within the bounds of the job assignment (Viscelli, 2016). Thus, the expected ability to control the truck makes the employee driver effectively an economic (but not legal) owner of the truck (Baker and Hubbard, 2004).

The multidimensional nature of EPRs implies that rights can be partitioned and distributed among different individuals or entities involved in business exchange with one another (Barzel and Allen, 2023: 22–24). Contracts are one means to distribute property rights among parties involved in exchange (Alchian, 1965; Cheung, 1970). For example, an employment contract conveys to a truck driver control over how to operate a company truck, with a trucking company voluntarily reducing the scope of its EPRs by surrendering some choices to a driver while, at the same time, retaining the right to determine the destination of the haul and how the truck is maintained (Ouellet, 1994). Thus, EPRs are often divided and distributed among various parties involved in production and exchange. Examples include shared ownership of intellectual property

developed in a joint venture or a rental contract for a car that conveys rights to an asset within restrictions stipulated in the contract (e.g., time, location, restricted uses) (Barzel and Allen, 2023).

The hypothesis at the core of property rights economics per Barzel and Allen (2023) is that the “*distribution of property rights will maximize the joint wealth net of transaction costs of the individuals involved [in the transaction]*” (65). Underlying this hypothesis is the assumption that individuals seek to maximize the economic value of their EPRs. They do this by (a) putting the resource into productive uses themselves or (b) ceding some or all their EPRs to another party if they are better equipped to generate economic value from the resource. An implication of this prediction is that the scope and strength of EPRs should increase with an individual’s ability to generate economic value from the resource. Thus, a partition of EPRs should reflect a situation in which parties engaged in exchange should be better off individually and collectively from the transfer of EPRs.

2.3 Proposed Modifications to PRTF

Despite its popularity, PRTF has been subject to numerous critiques (Demsetz, 1998; Foss and Foss, 2015; Kim and Mahoney, 2005) for (a) assuming asset ownership conveys residual control rights; (b) postulating that residual control rights are not easily divisible between multiple parties; and (c) treating enforcement by courts as perfect. Using theory elaboration (Fisher and Aguinis, 2017), and specifically Mäki’s (2004) principle of horizontal deisolation, we modify PRTF in two ways. Our first modification is introducing (Makadok et al., 2018) the distinction between LPRs and EPR into the PRTF framework. This distinction expands the domain of PRTF to any setting where an asset owner perceives some change in the external environment that reduces his/her expectation of the value he/she will obtain from consuming “the services of the asset [...] indirectly through exchange” (Barzel, 1994: 394) without altering LPRs *per se*. Thus, redefining PRTF in terms of EPRs brings many phenomena under a single umbrella (Mäki, 2001), a key theoretic virtue (Keas, 2018).

The second modification is replacing PRTF’s monolithic view of property rights with a multidimensional view. As stated by Kim and Mahoney (2005: 228), “ownership of an asset is not ownership of the physical entity itself, but ownership of specific property rights to an asset (or certain aspects of an asset).” This multidimensional view shifts the unit of analysis to the EPR per Coase (1960), as opposed to the asset (as in PRTF). By focusing on numerous EPRs associated with an asset, our modified PRTF is better able to explain intermediate forms of organizational design, such as settings in which a buyer owns an asset, yet the supplier uses the asset to deliver products/services to the buyer (Foss and Foss, 2015). Likewise recognizing multiple EPRs enables our modified version of PRTF to theorize why self-employed workers, upon having

some of their EPRs weakened (e.g., a reduction in the number of hours assets can be used in a day) may cede other EPRs (e.g., surrendering the ability to solicit business) to exchange partners (e.g., carriers better positioned to solicit hauls). This change thus further extends our modified PRTF's explanatory scope (Keas, 2018).

2.4 Property Right Restrictions Under Certainty-Based Enforcement Policies

Accounting, finance, and economics studies consider the firms' reporting and control environment as a determinant of misconduct, including misreporting of performance measures, criminal activity, and regulatory offenses, such as the HOS violations targeted by the ELD mandate (Campbell and Shang, 2022; Soltes, 2019). Firms' internal control systems rely on a mix of monitoring, incentives, and social controls to curb misconduct (Heese and Pérez-Cavazos, 2020; Liu, 2016; Maas and Van Rinsum, 2013; Raghunandan, 2021), while scrutiny from newspapers (Heese et al., 2022; Miller and Skinner, 2015), social media (Heese and Pacelli, 2024), and regulators (Defond and Lennox, 2017; Kedia and Rajgopal, 2011) may be deterrents external to the firm.

Our study is closely related to research on regulatory enforcement curbing misconduct (e.g., Polinsky and Shavell, 2000; Raghunandan, 2024; Raghunandan and Ruchti, 2024). A key tenet of the literature is that targeted individuals respond to certainty-based enforcement policies (Durlauf and Nagin, 2010), whereby social control agents make policy changes designed to increase actors' expected probabilities that misconduct will be detected and sanctioned (Becker, 1968; Chalfin and McCrary, 2017). Certainty-based enforcement policies may concentrate enforcement resources in a geographic area (e.g., hotspots policing) (Sherman and Rogan, 1995); among certain targets (e.g., guaranteed tax audits) (Kleven et al., 2011; Pomeranz, 2015); or during certain time periods (Eeckhout et al., 2010). Alternatively, as in our study, certainty-based enforcement policies may increase the expected probability of being detected by mandating electronic monitoring (Scott et al., 2021). Studies tend to find certainty-based policies deter misconduct (Chalfin and McCrary, 2017), albeit often fleetingly (Sherman, 1990).

Our research contributes to this stream of literature by extending the inquiry beyond the stated goal of curbing misconduct and shedding light on the potential adverse supply-side outcomes. Conceptualizing certainty-based enforcement policies as perceived restrictions in EPRs, we predict that reduced perceptions of operating flexibility can lead self-employed workers to reduce the labor they supply or even exit the market. Thus, our work alerts managers and regulators that rigorous enforcement may exacerbate labor shortages.

3 Institutional Background and Hypotheses

3.1 Electronic Logging Device Mandate in the For-Hire Trucking Industry

The for-hire truck transportation industry in the United States is massive, with employees and self-employed workers totaling over 1.8 million in 2022 (Bureau of Labor Statistics, 2023) and annual revenue of \$556 billion in 2022 (Census Bureau, 2022). The Federal Motor Carrier Safety Administration (FMCSA), a subdivision of the Department of Transportation, oversees motor carrier safety, a task of utmost public policy importance, given large truck crashes result in over 4000 deaths annually (FMCSA, 2022). One factor argued to contribute to these crashes is truck driver fatigue, which is often caused by truckers working excessively long hours (Ouellet, 1994). Consequently, the FMCSA establishes HOS rules that limit the number of hours truckers can work (Scott and Nyaga, 2019). One of the challenges of enforcing these rules is that until late 2017, truckers recorded their work hours using paper logbooks that were easy to falsify (Levy, 2023; Viscelli, 2016). Compounding the ease of falsifying work hours was the low probability of inspection, estimated at 0.2% on any given day for each driver (Balthrop et al., 2023). As a result, the industry was characterized by frequent violation of HOS rules (Levy, 2023). The FMCSA required most commercial trucks to be equipped with ELDs to curb HOS violations by making it more difficult—though not impossible (Levy, 2023)—for drivers to falsify their work hours. The ELD mandate was announced in 2015 and, despite many legal challenges, was implemented in two phases, with enforcement beginning in some states on 18 December 2017, and occurring in all states by 1 April 2018.

Prior studies (Levy, 2023; Miller et al., 2020; Scott et al., 2021) examine how a broad set of carriers responded to this change.² This paper builds on these studies by zeroing in on the behavior of FHOOs, which alleviates challenges associated with the delegation that exists within firms (Jensen and Meckling, 1976). This study also identifies a broader range of consequences of the ELD mandate—in that prior work has not examined impacts on labor supply, organizational form, and market exit. Furthermore, prior work only considered impacts after the ELD mandate was enforced, whereas this study observes behavior during the 2-year window between the announcement of the ELD mandate in December 2015 and its implementation (postannouncement period) to separately study the effects of the mandate on self-employed workers' operations during these two regulatory phases. Finally, our research design leverages the engine year carveout to define treatment and control groups within a narrow bandwidth of a few engine years around the cutoff with the intent to facilitate stronger causal identification.

3.2 Hypotheses

Consistent with the definition of EPRs under our modified PRTE, we view the start of the ELD mandate in December 2017 (postenforcement) as reducing the strength of FHOOs' perceived EPRs regarding how they can use their trucks (Barzel and Allen, 2023). Per the deterrence literature (Defond and Lennox, 2017; Kedia and Rajgopal, 2011), we assume that FHOOs required to install ELDs perceived an increased chance that, given inspection, noncompliance with HOS rules would be identified postenforcement. Qualitative research by Viscelli (2018) and Levy (2023) supports this assumption. Truckers have good reason to fear being found in violation of HOS rules as fines can reach \$16,000 and trigger a 34-hour suspension of operations (Scott et al., 2021). In our theoretical framing, a greater perceived probability of sanctioning for HOS violations constitutes a reduction in the strength of EPRs concerning how trucks can be operated while leaving LPRs associated with owning trucks unaffected. In contrast, FHOOs not affected by the ELD mandate could continue using paper logbooks, which afforded them continued flexibility to misreport work hours, thereby sustaining their EPRs (Levy, 2015). Due to the deterrent effect of stricter enforcement of HOS limits under mandatory ELDs, we expect treated FHOOs to engage in less misconduct as reflected in a reduced rate of major HOS violations in the postenforcement period.³

PREDICTION 1. Citations for major HOS violations decline among treated for-hire owner-operators after the ELD mandate takes effect (postenforcement).

The ability to transport lucrative hauls, often requested on short notice (Scott, 2015), is an EPR of high operational value to FHOOs (Barzel and Allen, 2023: 73). Prior to the ELD mandate, paper logbooks gave FHOOs flexibility to accommodate unforeseen challenges on the road, such as delays during pick-up or inclement weather (Maister, 1980). Paper logbook protected these EPRs and provided considerable economic value to FHOOs who are residual claimants to the profits generated from their assets (trucks) (Barzel and Allen, 2023: 17). For example, Scott and Nyaga (2019) report that, prior to the ELD mandate's enforcement, FHOOs' propensity to commit major HOS violations increased with spot market trucking prices, consistent with the industry's "get while the getting is good" mentality (Ouellet, 1994).

Stricter enforcement of HOS rules due to the ELD mandate restricts FHOOs' expected ability to complete hauls by working beyond HOS limits. Hence, FHOOs required to install ELDs become less likely to accept hauls that require driving in violation of HOS rules. This curtailment to respond to unforeseen opportunities and challenges depresses FHOOs' expectations about the productive potential of their trucks. We conceptualize this loss in operating flexibility as a reduction in the strength of EPRs (Barzel and Allen, 2023; Coase, 1960). In response to reduced operating flexibility, treated FHOOs should be less willing to accept hauls that risk violating HOS

limits if completed on time, thereby scaling back their supply of labor. In contrast, their counterparts operating trucks exempt from ELDs experience no change in their EPRs and will not reduce their supply of labor. As with major HOS violations, we expect that labor supply reductions will only occur during the postenforcement phase, as FHOOs can quickly adjust operations in response to inspection activity (Balthrop et al., 2023).

PREDICTION 2. Volume of labor supplied declines among treated for-hire owner-operators after the ELD mandate takes effect (postenforcement).

Under our modified PRTE, EPRs associated with asset ownership are not unidimensional, but rather a bundle of property rights that can be partitioned (Alchian and Demsetz, 1973; Demsetz, 1967). If EPRs with respect to an asset are divided between several parties, all parties holding EPRs exercise arguably some level of "economic ownership" over an asset. For example, Barzel and Allen (2023: 26) refer to a secretary as an economic but not a legal owner of an organization's copy machine. The partitioning of EPRs has ramifications for organizational design choices. In our setting, self-employed truckers choose between either holding most EPRs with respect to their trucks and operating independently or ceding some of their EPRs by leasing themselves as subcontractors to motor carriers for a period (e.g., 6 months).

A subcontracting relationship is broadly defined as an arrangement under which a firm procures a product or service that it is normally capable of producing (Day, 1956; van Mieghem, 1999). Subcontracted owner-operators benefit from the carriers' relationships with shippers, which allow carriers to provide subcontractors with superior load matches that can improve revenue productivity (Li et al., 2022). Rakowski (1988) labels that operating efficiencies carriers attain by pooling large volumes of freight from multiple shippers "marketing economies," a term that we adopt because it generalizes to other settings (e.g., crowdsourcing). Although a subcontractor surrenders his ability to solicit his own (Maister, 1980; Viscelli, 2016), he continues to hold the right to refuse offered shipments (Baker and Hubbard, 2004) and is largely free from day-to-day control (Miller et al., 2018) unlike employee truckers who must accept the loads that their carriers assign to them.

From a self-employed driver's perspective, the decision to subcontract in lieu of independent operations reflects a trade-off between retaining EPRs of freight solicitation and sole residual claimant status and gaining the ability to leverage a larger carrier's marketing economies. We predict that EPR restrictions imposed by the ELD mandate will reshape this trade-off and push FHOOs toward more subcontracting. Before being required to use ELDs, FHOOs could feasibly work up to 16 or 17 hours a day well beyond the 14-hour limit, which permits driving more miles and hence increasing earnings. As discussed under our second prediction,

this flexibility allowed FHOOs to accept highly profitable spot market truckload shipments before the ELD mandate diminished upon the mandate's introduction. Reduced EPRs associated with limited operating flexibility (i.e., the ability to accept and use their truck within a specific time frame) will diminish the economic value of complementary EPRs associated with soliciting business (Barzel and Allen, 2023: 73). We predict that this loss of EPRs (operating flexibility) spurs a reallocation of complementary EPRs (freight solicitation) into the hands of larger carriers that can generate a comparatively larger economic value from the EPR under the changed operating environment.

Our prediction of increased subcontracting aligns with Barzel and Allen's (2023) property rights hypothesis that a given EPR will be assigned to the party with the largest influence on the income flow from that EPR (Barzel and Allen, 2023: 66–70; Libecap, 2003; Mahoney, 2004). Notably, it also requires a multidimensional conceptualization of EPRs that modern PRTF emphasis on asset ownership fails to recognize. As it takes time for FHOOs to find suitable carriers to subcontract with, we expect FHOOs to commence this search in advance of the ELD mandate's enforcement. Thus, unlike the first two predictions, the third anticipates a change in treated FHOOs' behavior beginning after the ELD mandate is announced and continuing into the postenforcement period:

PREDICTION 3. *Reliance on subcontracting increases among treated for-hire owner-operators after the ELD mandate is announced (postannouncement) and takes effect (postenforcement).*

While we cannot directly observe the marketing economies mechanism underlying our third prediction, examining whether treated FHOOs are more likely to subcontract with larger than smaller motor carriers may provide indirect evidence. Larger motor carriers are more likely to be invited to large shippers' strategic procurement auctions—events where shippers award contracted freight volumes to carriers (Caplice, 2007)—because larger carriers can offer a shipper greater capacity than smaller carriers, which minimizes the transaction costs shippers incur (Li et al., 2022). Thus, larger motor carriers should have greater marketing economies (Rakowski, 1988) because, as with pooling demand when centralizing inventories (Eppen, 1979), larger size results in less volatility of demand relative to average demand (Whitin and Peston, 1954). Given the heavily random nature of shippers' demand for truckload transportation (Powell et al., 1988), larger carriers that offer smoother demand patterns will appear more attractive to FHOOs after the ELD mandate diminished the EPRs associated with the hours they can use their trucks:

PREDICTION 4. *Reliance on subcontracting among treated for-hire owner-operators after the ELD mandate is announced*

(postannouncement) and takes effect (postenforcement) increases more with larger carriers than smaller carriers.

As EPRs shape the behavior of those individuals that hold them, our modified PRTF lends itself to the examination of another decision within self-employed workers' opportunity set (Foss and Foss, 2022): market exit. Our modified PRTF assumes that individuals seek to maximize the economic value generated by their assets. A reduction in operating flexibility under the ELD mandate and the concomitant decline in the strength of EPRs depresses treated FHOOs' expectations of generating positive income flow from their assets (Barzel and Allen, 2023). Findings from economics suggest that firms with low productivity are far more likely to cease operations (Syverson, 2011). Exit rates are sensitive to declines in expected productivity in our setting because many FHOOs have limited financial resources to withstand operating losses (Cantor et al., 2013). A negative shock to EPR strength could cast doubts for some treated FHOOs over the viability of their business as going concern. The prediction provides for a direct test of the complementarity between EPRs of operating flexibility over the asset and EPRs to the residuals that motivate a self-employed worker to maximize the income flow from his asset (Foss and Foss, 2022). Also, among a subset of treated FHOOs who are morally opposed to ELD monitoring (Levy, 2023), the ELD mandate may have diminished the perceived net utility of continuing to operate for reasons independent of productivity concerns, such as diminishing job satisfaction. Both rationales align with our modified version of PRTF, which further illustrates the benefit of foregrounding EPRs in theoretical models of the firm. As with subcontracting, a decision to exit takes time to implement. Therefore, we expect elevated market exit by treated FHOOs after the ELD mandate is announced and before it takes effect:

PREDICTION 5. *Market exits increase among treated for-hire owner-operators after the ELD mandate is announced (postannouncement) and takes effect (postenforcement).*

4 Data, Research Design, and Sample

To test our predictions regarding the effects of the ELD mandate on FHOOs, we use data contained in the Motor Carrier Management Information System (MCMIS) collected by the Federal Motor Carriers Safety Administration (FMCSA) between 2012 and 2020. MCMIS census files include firm-level information about the number of drivers employed and subcontracted and the number of trucks owned and leased by each carrier. MCMIS inspection files document inspections of trucks along roadsides and at fixed inspection stations, providing a representative snapshot of driver behaviors—including violations and subcontracting. We use Department of Transportation (DOT) numbers⁴ to match inspected trucks to the carriers under whose authority they are operating at inspection. Given the focus of our study on FHOOs, we limit our

analysis to vehicles inspected between 2012 and 2019 operating under DOT numbers that report only one owned truck in the census files.

Our DiD research design relies on a regulatory exemption for trucks with engine years before the year 2000 to construct a control group of drivers unaffected by the ELD regulation. The advantage of such a DiD design is that the control group provides a counterfactual estimate of outcomes for FHOOs in the absence of treatment by the ELD regulation against which the outcomes of the treatment group can be assessed (Angrist and Pischke, 2009: 52; Morgan and Winship, 2015: 37; Roberts and Whited, 2013: 42). Utilizing a control group effectively mitigates concerns that demand-side factors such as general trends in demand for truckload transportation or changes in service prices that may coincide with the introduction of the ELD mandate and also affect FHOOs' outcomes threaten our inferences. The key identifying assumption of a DiD design is the presence of parallel preperiod trends for the respective outcome between the treated and control groups that substantiates that both groups are subject to the same shocks (Roberts and Whited, 2013: 41–43). We assess the validity of the parallel trends assumption by graphically plotting estimated quarterly treatment effects before treatment (Angrist and Pischke, 2009: 177–180; Armstrong et al., 2022: 23–26; Atanasov and Black, 2016: 241–243). The plots as well as formal tests do not reveal any divergent trends in the preperiod leading up to the introduction of the ELD mandate, which suggests that labor outcomes for drivers in the treated and control groups move in parallel through time due to being exposed to the same economic shocks.

We extract model years from vehicle identification numbers (VINs) to segregate trucks subject to the ELD mandate from those exempted under the carveout. To keep treatment and control groups as comparable as possible, we restrict our main analysis to narrow bandwidths of trucks manufactured 3 years before (1997–1999) and 3 years after (2001–2003) the carveout cutoff model year.⁵ Excluding trucks with inspections under non-US DOT numbers reduces our sample to 1,468,530 inspections and 199,246 unique trucks operated by FHOOs. We report descriptive statistics for FHOO behavior between January 2012 and the announcement of the ELD mandate in December 2015 (the preperiod) for both the treatment and control group in Table EC2 in the E-Companion. In the preperiod, FHOOs driving vehicles later subject to the mandate (treated) were inspected (as captured by the variable *drive*) on average every 9 months or with a probability of 10.8% each month. These drivers relied more heavily on independent operations than on subcontracting⁶ (only 17.6% of inspections) and were cited for major HOS violations⁷ in 3.5%, logbook violations in 8.3%, and minor HOS violations in 8.8% of inspections.

Subsample comparisons for the descriptive statistics reported in Table EC2 reveal that the drivers in the treatment group of our main analysis with slightly newer trucks (12.4 vs. 16.1 years) tend to *drive* more (10.8% vs. 10.1% year-month probability of inspection), *exit* less (0.3% vs. 0.4%

year-month probability), commit more *major HOS violations* (3.5% vs. 2.9% year-month probability), and are more likely to *subcontract* (17.6% vs. 15%), although the disparities tend to be moderate in magnitude. To further minimize covariate imbalance between treatment and control group, we limit the differences along the forcing variable (i.e., engine age) by restricting the sample to the smallest possible increment of 1 year before (1999) and 1 year after (2001) the carveout cutoff model year. This approach further reduces the average age difference between treatment and control group from 3.7 years to only 2.0 years. As expected, differences in covariate means further decrease as the bandwidth narrows from three model years to one model year (see Table EC2, columns 6 vs. column 3). Akin to a regression discontinuity design (Imbens and Lemieux, 2008), the sample restriction bolsters the credibility of our identification strategy as treatment and control group are highly similar except for minimal differences in the forcing variable and its potential correlates.⁸ Our inferences for the restricted sample analysis generally yield unchanged inferences and are reported in a separate column throughout all our tables.⁹

For our tests, we employ a specification that differentiates between effects in the postannouncement and posteffective period. Our resulting three-period DiD model captures changes in behavior after the ELD mandate announcement (December 16, 2015–December 17, 2017) and effective dates (December 18, 2017–December 31, 2019) separately using the 3 years before the ELD mandate announcement as a benchmark period (January 1, 2012–December 15, 2015). This approach allows us to estimate anticipation effects that occur in response to the mandate announcement but before the mandate takes effect.

5 Findings

5.1 Deterrence of Misconduct

Prediction 1 posits a deterrent effect consistent with the intent of the ELD mandate that strengthening enforcement will curb misconduct. To test this prediction, we explore changes in major HOS violations (e.g., driving more than 11 hours in a 14-hour window). An observed decrease in citations for major HOS violations among treated for FHOOs requires that the deterrence mechanism overpowers any improvement in detection facilitated by the enforced adoption of ELDs after the mandate (Scott et al., 2021). Comparing changes in citations for major HOS violations is, therefore, a conservative test. Using inspection-level data and OLS regressions, we estimate the following DiD linear probability model (LPM):

$$\begin{aligned} \text{prob}(\text{major HOS violation}_{it} = 1) \\ = \beta_0 + \beta_1 \text{post_announce}_t \times \text{treat}_i + \beta_2 \text{post_effective}_t \\ \times \text{treat}_i + \alpha_i + \delta_t + \epsilon_j \end{aligned} \quad (1)$$

Subscript i denotes the vehicle, and t denotes the year-month. *Major HOS violation* is an indicator variable, indicating a

Table 1. The deterrence effect of the ELD mandate on owner-operator misconduct.

Sample Dependent variable	± 3 model years		± 1 model year	± 3 model years	± 3 model years
	Major HOS violation			Major log violation	Minor HOS violation
	(1)	(2)	(3)	(4)	(5)
Post_announce $\times$ treat	-0.0007 (0.0007)	0.0012 (0.0008)	-0.0008 (0.0011)	-0.0008 (0.0012)	-0.0003 (0.0013)
Post_effective $\times$ treat	-0.0048*** (0.0007)	-0.0028*** (0.0008)	-0.0035*** (0.0011)	-0.0055*** (0.0014)	0.0133*** (0.0014)
State inspection intensity		-0.0083*** (0.0002)			
Lagged accident		0.0021 (0.0021)			
Lagged drive		-0.0002 (0.0005)			
Lagged maintenance violation		-0.0001 (0.0005)			
Lagged traffic violation		0.0017*** (0.0005)			
Panel fixed effects	Truck	Truck	Truck	Truck	Truck
Time fixed effects	Model years	Model years	Model years	Model years	Model years
N (inspections)	1,468,530	1,192,490	622,163	1,468,530	1,468,530
Adjusted R ²	0.0328	0.0366	0.0323	0.0844	0.0719
$F(\beta_1 - \beta_2 = 0)$	29.16***	26.57***	4.99**	10.31***	78.07***

Notes. Robust standard errors clustered by truck are reported in parentheses. We compare postannouncement with posteffective treatment effects in the final row. See Table EC1 in the E-Companion for variable definitions. *, **, and *** indicate significance at the 10%, 5%, and 1% levels (two-tailed).

citation for a major HOS violation during an inspection. We specify two-way fixed effects using vehicle (α) fixed effects to control for time-invariant differences between treatment and control vehicles and month-year (δ) fixed effects to control for common trends, such as changing demand for truck transportation services (Bertrand et al., 2004). *Treat* is an indicator variable equal to one for trucks in the treatment group, and *postannounce* (*posteffective*) is a period indicator variable for months after the announcement date but before the effective date (after the effective date). β_1 and β_2 estimate the ELD mandate's treatment effects as changes in the probabilities of receiving a major HOS violation conditional upon inspection for treated FHOOs relative to the control group. β_1 captures anticipatory effects during the announcement period, and β_2 captures the effects of the mandate after the effective date. All inferences are based on standard errors clustered by truck.¹⁰ We provide a list of definitions for all variables used in our analyses in Table EC1 in the E-Companion.

Consistent with prediction 1, we find evidence that the ELD mandate has a deterrence effect. Column 1 of Table 1 reveals a 0.48% point decrease ($p < .01$) in the probability of *major HOS violations* given inspection among treated FHOOs after the effective date. Our results are robust to the inclusion of controls for for-hire owner-operator behavior in the 12 months preceding inspection, such as traffic or maintenance violations, accidents, and driving behavior as well as

for inspection intensity in the inspection state (column 2). The results persist when we limit our sample to vehicles in a 1-year bandwidth around the carveout cutoff (column 3).¹¹ Figure 1(a) plots the quarterly leads and lags of this effect and depicts a sharp decrease in the incidence of major HOS violations among treated FHOOs after the mandate effective date, substantiating that the ELDs deter misconduct. Importantly, the absence of a discernable trend in the estimated treatment effects in the period leading up to the mandate announcement (all statistically insignificant) provides evidence supporting the parallel trends assumption underlying our DiD research design. Relative to preannouncement levels for the treated group, the reported coefficient reflects a decrease in *major HOS violations* ranging from 9.3% and 13.7%, depending on the specification.¹² Using citations for the detected manipulations of paper or electronic record of duty status logs to conceal HOS violations as an alternative measure of misconduct targeted by the mandate (column 4), we found a significant 0.55% point decrease ($p < .01$) in the probability of a violation, which is a 6.6% decrease in the incidence of log violations over preperiod levels.

While *major HOS violations* and *log violations* may reflect deliberate efforts to drive more than HOS limits permit, *minor HOS violations* capture clerical and technical errors, including failure to retain the ELD user manual and incorrect setup of the ELD device. Therefore, the more rigorous enforcement of HOS limits implemented by the ELD mandate and

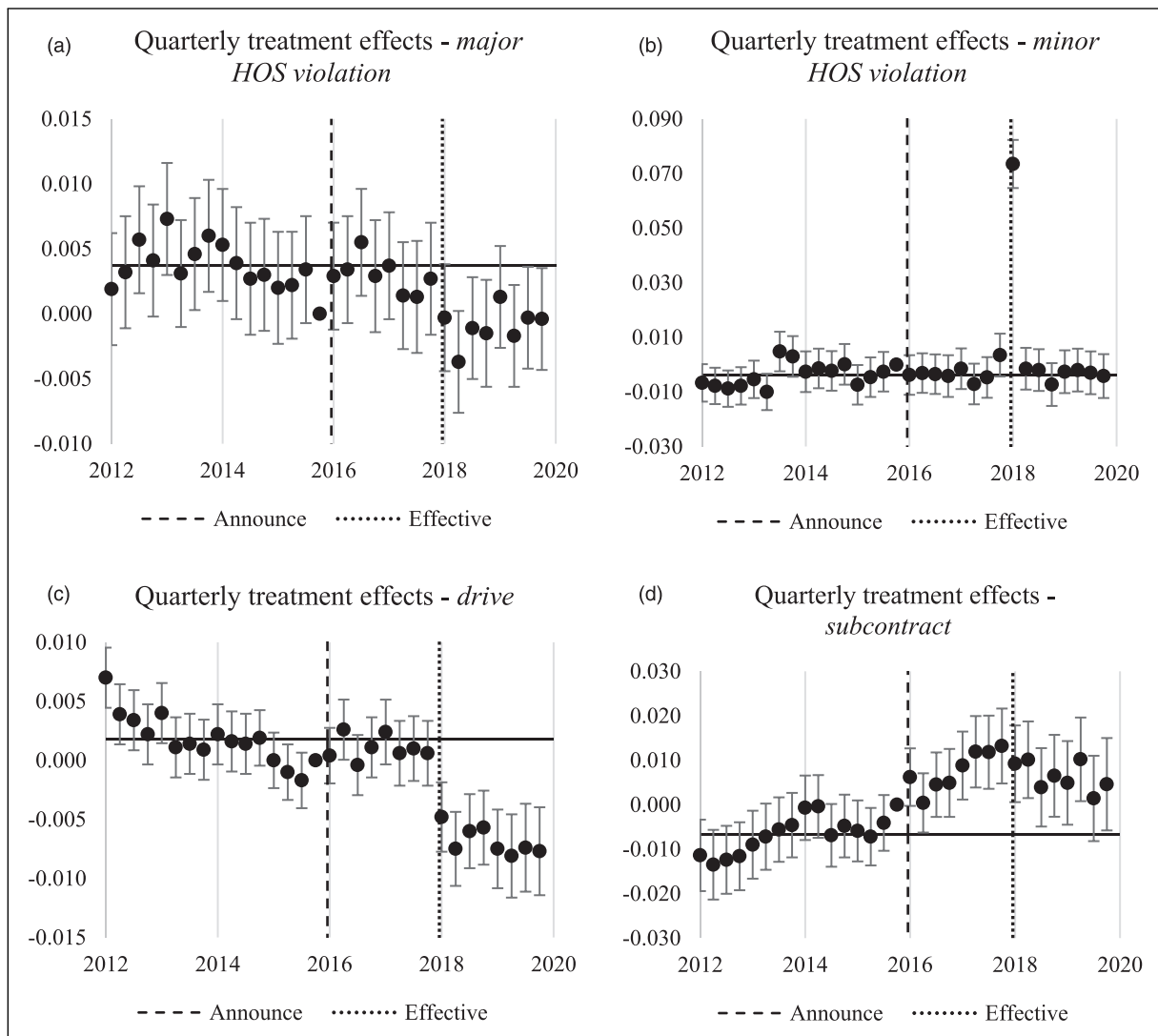


Figure 1. Treatment Effects of the ELD mandate on owner-operator behavior.

Notes: Panels (a–d) present the quarterly effects of treatment on key dependent variables for the three-model-year treated group relative to the corresponding control group. The effects in panel (c) are estimated using truck-month-year observations. Remaining figures use inspection-level observations. All figures rely on month-year and truck fixed effects. 95% confidence intervals are calculated using robust standard errors clustered by truck. The baseline represents the average treatment effect estimated for the preperiod (i.e., all quarters between January 2012 and the ELD mandate announcement in December 2015). The ELD mandate announcement and effective dates are marked.

drivers' learning over time how to install and operate ELDs correctly should manifest as a temporary increase in citations for minor HOS violations among treated FHOOs after the mandate went into effect. We found a 1.33% point increase (column 5, $p < .01$) in the probability of a minor HOS violation given inspection, corresponding to a 15% increase in *minor HOS violations* among vehicles subject to the ELD mandate. The spike in *minor HOS violations* among treated FHOOs right after the mandate took effect, depicted in Figure 1(b), corroborates the installation of ELDs among vehicles subject to the mandate. This further eliminates the argument by Levy

(2023) that inspectors may have less stringently examined HOS compliance once the ELD mandate started.

5.2 Volume of Labor Supplied

Prediction 2 maintains that reduced operating flexibility under the ELD mandate diminishes treated FHOOs' productivity and thus their incentives to supply labor. To test this, we use the occurrence of inspections to gauge the volume of labor supplied following Scott et al. (2021). Accordingly, the probability of inspection should fall as treated FHOOs accept fewer

Table 2. The effect of the ELD mandate on labor supplied.

Sample	±3 model years	±3 model years	±1 model year
Dependent variable	Drive		
	(1)	(2)	(3)
Post_announce × treat	−0.0006 (0.0006)	−0.0004 (0.0006)	0.0009 (0.0009)
Post_effective × treat	−0.0085*** (0.0008)	−0.0078*** (0.0008)	−0.0066*** (0.0012)
State inspection intensity		0.0033*** (0.0002)	
Lagged accident		−0.0168*** (0.0017)	
Lagged maintenance violation		0.0204*** (0.0003)	
Lagged traffic violation		0.0082*** (0.0005)	
Panel fixed effects	Truck	Truck	Truck
Time fixed effects	Model years	Model years	Model years
N. (truck × month-years)	12,785,980	9,919,508	5,414,403
Adjusted R-squared	0.0783	0.0792	0.0770
F ($\beta_1 - \beta_2 = 0$)	113.87***	101.34***	42.57***

Notes. Robust standard errors clustered by truck are reported in parentheses. We compare postan-
nounce with posteffective treatment effects in the final row. See Table EC1 in the E-Companion for variable
definitions. *, **, and *** indicate significance at the 10%, 5%, and 1% levels (two-tailed).

hauls and spend fewer hours on the road. We use *drive* to indicate whether an owner-operator's truck was inspected in a month-year:

$$\begin{aligned}
 \text{prob}(\text{drive}_{it} = 1) \\
 = \beta_0 + \beta_1 \text{post_announce}_i \times \text{treat}_i + \beta_2 \text{post_effective}_i \\
 \times \text{treat}_i + \alpha_i + \delta_t + \varepsilon_{it}
 \end{aligned} \quad (2)$$

Consistent with prediction 2, Table 2, column 1 shows a significant 0.85% point decrease ($p < .01$) in the probability of observing an inspection for treated FHOOs each month after the mandate effective date.¹³ The decrease in labor supplied manifests as a drop in *drive* among treated FHOOs after the mandate effective date as shown in Figure 1(c). The absence of discernable trends in estimated treatment effects leading up to the mandate announcement, most of which are statistically insignificant, is consistent with the assumption of parallel trends between the treatment and control groups underlying our DiD design. To address concerns that changes in enforcement after the mandate may explain decreases in the frequency of inspection, we control for the model-year-specific frequency of inspection for the state of inspection per year-month (Table 2, column 2) and continue to find significant results.¹⁴ Our results are also robust to using the narrower bandwidth of trucks manufactured 1 year around the legacy cutoff (Table 2, column 3). The observed effect sizes reflect a decrease in the probability of inspection among treated for-hire owner-operators ranging between 6.2% and 7.9%.¹⁵ Thus,

we found evidence that the ELD mandate taking effect reduced treated FHOO volume of labor supplied.

5.3 Subcontracting

Prediction 3 posits that treated FHOOs will shift toward subcontracting. We operationalize *subcontract* to indicate an inspection under a DOT number assigned to a carrier that relies significantly (at least 80%) on subcontracted drivers as reported in the census. To test prediction 3, we use inspection-level observations and replace *subcontract* as the dependent variable in equation (1). Results in column 1 of Table 3 reveal that treated FHOOs experience a 1.38% and 1.35% point increase in the probability of subcontracting given inspection after the announcement and effective dates, respectively ($p < .01$). Increases in subcontracting after the announcement and effective dates remain significant ($p < .01$) when we narrow the bandwidth of model years considered (Column 2) and when we include a full set of controls (untabulated). Based on preperiod treatment group subcontracting levels, these effects constitute a 7.8%–8.8% range increase in subcontracting after the mandate announcement and a 7.6%–7.9% range increase after the effective date. The postannouncement increase in subcontracting is also evidenced in Figure 1(d), which displays quarterly treatment effects for *subcontract*. Figure 1(d) does not show any discernable treatment effect leading up to the mandate announcement and none of the estimated treatment effects before mandate announcement are significant, again validating the parallel trends assumption.

Table 3. The effect of the ELD mandate on subcontracting.

Sample	± 3 model years		± 1 model year	
Dependent variable	Subcontracting		Subcontracting with small carriers (5–12 vehicles)	
	(1)	(2)	(3)	(4)
Post_announce $\times$ treat	0.0138*** (0.0021)	0.0151*** (0.0033)	0.0038*** (0.0010)	0.0026* (0.0016)
Post_effective $\times$ treat	0.0135*** (0.0027)	0.0133*** (0.0042)	0.0012 (0.0013)	−0.0013 (0.0020)
Panel fixed effects	Truck	Truck	Truck	Truck
Time fixed effects	Model years	Model years	Model years	Model years
N (inspections)	1,468,530	622,163	1,468,530	622,163
Adjusted R ²	0.6332	0.6348	0.5604	0.5426
$F(\beta_1 - \beta_2 = 0)$	0.02	0.26	4.99**	4.63**
Subsample	± 3 model years		± 1 model year	
Dependent variable	Subcontracting with medium carriers (13–53 vehicles)		Subcontracting with large carriers (54+ vehicles)	
	(5)	(6)	(7)	(8)
Post_announce $\times$ treat	0.0020* (0.0012)	0.0030 (0.0018)	0.0073*** (0.0015)	0.0095*** (0.0024)
Post_effective $\times$ treat	0.0029** (0.0015)	0.0044* (0.0023)	0.0096*** (0.0020)	0.0102*** (0.0032)
Panel fixed effects	Truck	Truck	Truck	Truck
Time fixed effects	Model years	Model years	Model years	Model years
N (inspections)	1,468,530	622,163	1,468,530	622,163
Adjusted R ²	0.5878	0.5919	0.6538	0.6558
$F(\beta_1 - \beta_2 = 0)$	0.52	0.51	1.80	0.07

Notes. We compare postannouncement with posteffective treatment effects in the final row. Robust standard errors clustered by truck are reported in parentheses. See Table EC1 in the E-Companion for variable definitions. *, **, and *** indicate significance at the 10%, 5%, and 1% levels (two-tailed).

To test prediction 4 concerning differences in the change in the incidence of subcontracting with large and small carriers, we examine which carrier segment FHOOs subcontract by partitioning carriers into terciles by fleet size: small carriers (fleet size 5–13 vehicles), medium carriers (fleet size 14–54 vehicles), and large carriers (fleet size >54 vehicles). We redefine our dependent variable subcontract as an inspection under a DOT number associated with a carrier in each size tercile that relies on at least 80% subcontracted drivers (Table 3, columns 3 through 8). Letting k index each version of the dependent variable across the three size classes, we stack the three dependent variables and estimate a panel data seemingly unrelated regression (Miller et al., 2022). We include truck-by-dependent variable (α_{ik}) and month-year-by-dependent variable set fixed effects (δ_{tk}) and cluster standard errors by vehicle.

$$\begin{aligned}
 \text{prob}(\text{subcontracting}_{itk} = 1) \\
 = \beta_0 + \beta_1^k \text{post_announce}_{itk} \times \text{treat}_{itk} + \beta_1^k \text{post_effective}_{itk} \\
 \times \text{treat}_{itk} + \alpha_{ik} + \delta_{tk} + \varepsilon_{itk}
 \end{aligned} \quad (3)$$

The results from this procedure¹⁶ are reported in Table 3, columns 3–8. Focusing on the 3-year window, the shift toward subcontracting among treated FHOOs after announcement to large carriers is 0.73% points ($\beta_1^{\text{Large}} = 0.0073$ from column 7) versus 0.38% points for small carriers ($\beta_1^{\text{Small}} = 0.0038$ from column 3). Constraining these parameters to equality results in a significant Wald test ($\Delta F = 14.58$, $p < .01$). We reach a similar conclusion with the 1-year sample ($\beta_1^{\text{Large}} = 0.0095$ from column 8 vs. $\beta_1^{\text{Small}} = 0.0026$ from column 4; $\Delta F = 23.29$, $p < .01$). Likewise, the shift toward subcontracting among treated FHOOs after enforcement to large carriers is 0.96% points ($\beta_2^{\text{Large}} = 0.0096$ from column 7) versus 0.12% points to small carriers ($\beta_2^{\text{Small}} = 0.0012$ from column 3). Constraining these parameters to equality results in a significant Wald test ($\Delta F = 62.82$, $p < .01$). We reach a similar conclusion with the 1-year sample ($\beta_2^{\text{Large}} = 0.0102$ from column 8 vs. $\beta_2^{\text{Small}} = -0.0013$ from column 4; $\Delta F = 49.34$, $p < .01$). Therefore, increased subcontracting to the largest tercile of carriers accounts for most of the shift toward subcontracting. This aligns with our explanation that treated FHOOs subcontract to leverage carriers' marketing economies.

5.4 Market Exit

Prediction 5 posits that treated FHOOs will be more likely to exit the market. The absence of future inspections allows us to infer FHOOs' market *exit*, which we define on a truck-month-year basis. To adjust for any driving after the last observed inspection, we estimate market *exit* as a lagged event, occurring with the truck-specific mean number of months between past inspections after the final inspection.^{17,18} We estimate the monthly probability of market *exit* using a Cox proportional-hazards model with time-varying covariates (Clotfelter et al., 2008; Cox, 1972; Kleyменова and Tomy, 2022; Wooldridge, 2015).¹⁹ The model predicts the hazard of *exit* h at a month-year t , given that an owner-operator i has not yet exited:

$$h(m) = h_0 \times (\beta_1 \text{post_announce}_t \times \text{treat}_i + \beta_2 \text{post_effective}_t \times \text{treat}_i + \beta_3 \text{treat}_i + \beta_4 \text{vehicle age}_{it}) \quad (4)$$

Above, m measures the number of months for year-month t since the start of sample period (2012). A significant β_1 or β_2 rejects the proportional hazard assumption with respect to *treat*, with a positive effect suggesting that treated FHOOs experience an increase in the hazard of *exit* after the introduction of the mandate. We include *vehicle age*, a time-varying covariate, in our proportional-hazards model to account for any change in the hazard of *exit* associated with aging of their vehicles.

The average age of vehicles included in our study at the beginning of our sample period is 12.9 years (untabulated). Despite this advanced age, *exit* was infrequent for all vehicles with a probability of less than 1% per month in the preperiod (see Table EC2). Consistent with prediction 5, we found an increase in the hazard of *exit* (Table 4, column 1, $p < .01$) among treated FHOOs after the effective date (*post-effective*).^{20,21} Additionally, our estimates show significant increases in *exit* after announcement ($p < .01$), suggesting some FHOOs may accelerate their market *exit* decision in anticipation of looming reductions in EPRs. The finding persists when we include a full set of controls (Table 4, column 2) and is similar if we restrict the vehicles to those within a 1-year bandwidth around the legacy cutoff (Table 4, column 3). Depending on the specification, these estimates imply a 3.6%–12.5% increase in market exit after the announcement and a 11.1%–21.6% increase in market exit after the effective date.

When FHOOs abandon independent operations, they may either choose alternative work arrangements, such as subcontracting (Prediction 3), or exiting the industry entirely. When operating independently, FHOOs drive under their own DOT number. Incorporating that FHOOs may abandon their own DOT number to continue to subcontract, we define a broader measure of exit—*exit from independent driving*—as occurring when the last year-month a vehicle is inspected under a DOT number belonging to an IC. The results of this alternative measure of market exit reported in column 4 document

a significant 4.63% increase in the probability of permanent abandonment of independent work ($p < .01$) after the mandate effective date.

6 Discussion

6.1 Theoretical Contributions

The first way this manuscript contributes to theory is by introducing the distinction between EPRs and LPRs into PRTF. Modern PRTF's focus on asset ownership per se limits itself to LPRs, which is problematic for two reasons. First, strong evidence exists that EPRs, not LPRs, drive individual behavior (Barzel, 1994; Foss and Foss, 2015). Second, modern PRTF's focus on LPRs limits its scope of applicability, in that it can only be used in settings in which shifts occur in LPRs (Foss and Foss, 2022). In this research, we follow Makadok et al. (2018) in adopting a different mode of theorizing to modify PRTF to explicitly focus on how shifts in EPRs shape actors' behaviors. In doing so, our modified version of PRTF is now applicable to settings in which regulatory shifts change actors' perceptions of their EPRs even when LPRs are not impacted, which may provide a basis for better understanding how self-employed workers respond to regulatory actions that restrict their EPRs. Moreover, there are numerous applications for our EPR-focused PRTF beyond the impact of regulations on self-employed workers. For example, this theory can explain why some of the largest US importers, such as Walmart, The Home Depot, and Costco, chartered containerships in 2021 to deliver time-critical holiday products. Long vessel queues at US container ports coupled with frequent cargo rolls—when ocean carriers shift freight onto another vessel leaving at a later date—reduced these importers' perceptions of the net value they would obtain from the ocean carriers' vessels via market exchange. Chartering their own ships, although costly, strengthened their EPRs (Baertlein et al., 2021). Likewise, our EPR focused PRTF can explain why some retailers are closing outlets in urban areas: heightened rates of rampant theft reduce perceived expected profitability from store operations (Foss and Foss, 2022). The ability of our extended PRTF to unite a wide range of phenomena is a key theoretic virtue (Mäki, 2001; Thagard, 1978).

The second way this manuscript contributes to theory is by replacing modern PRTF's unidimensional conceptualization of property rights that are tied explicitly to asset ownership per se with a multidimensional view of EPRs that encompasses the various uses that EPR holders can undertake with their assets (Coase, 1960; Kim and Mahoney, 2005). One conceptual improvement (Laudan, 1977) over modern PRTF our modified theory offers is to better account for intermediate forms of organization. For example, subcontracting does not fit well with canonical PRTF because the subcontractor owns the productive assets yet does not have the same flexibility that comes with truly independent operations (Day, 1956; Maister, 1980; van Mieghem, 1999). Our modified PRTF accommodates subcontracting by recognizing that it involves surrendering EPRs

Table 4. The effect of the ELD mandate on market exit.

Sample	±3 model years	±3 model years	±1 model year	±3 model years
Dependent variable	Hazard of exit			Hazard of exit from independent driving
	(1)	(2)	(3)	(4)
Post_announce × treat	0.1023*** (0.0176)	0.1181*** (0.0184)	0.0360 (0.0268)	0.0315* (0.0177)
Post_effective × treat	0.1960*** (0.0162)	0.2256*** (0.0171)	0.1055*** (0.0248)	0.0453*** (0.0172)
Treat	−0.0024 (0.0195)	−0.0451** (0.0204)	0.6556 (0.0000)	0.0128 (0.0197)
Vehicle age	0.0217*** (0.0041)	0.0250*** (0.0042)	0.3221*** (0.0090)	0.0181*** (0.0043)
State inspection intensity		−0.0899*** (0.0032)		
Lagged accident		0.0460 (0.0361)		
Lagged drive		0.3201*** (0.0112)		
Lagged maintenance violation		0.3116*** (0.0098)		
Lagged traffic violation		0.0205** (0.0099)		
N (truck × month-years)	15,996,894	8,499,905	6,780,432	15,865,345
Wald χ^2	182.61***	6648.87***	3225.86***	46.11***
χ^2 ($\beta_1 - \beta_2 = 0$)	29.95***	39.28***	7.04***	0.52

Notes. This table reports the effect of the ELD mandate on the hazard of market exit (columns 1–3) and the hazard of exit from independent operations (column 4) obtained from estimating the Cox proportional hazards model presented in equation (4). Vehicle age is a continuous, time-varying measure. All other control variables are time-varying indicators. We compare postannounce with posteffective treatment effects in the final row. Robust standard errors clustered by truck are reported in parentheses. See Table ECI in the E-Companion for variable definitions. *, **, and *** indicate significance at the 10%, 5%, and 1% levels (two-tailed).

associated with soliciting independent business. This is not unique to our trucking context: for example, drivers working for ridesharing companies such as Uber are precluded from soliciting business from the users outside of the app (Uber, 2023). Likewise, our modified PRTF can be used to explain scenarios where buyers own equipment that is utilized by suppliers (Rossetti and Choi, 2005), in that some EPRs belong to the buyer (e.g., the right to remove equipment from a supplier's plant) while others belong to the supplier (e.g., the right to residual profits). Thus, this modification of PRTF not only addresses findings that were previously anomalies to the theory—recognized as one of the strongest types of theoretical contributions (Laudan, 1977)—our modified PRTF has the ability to unify the understanding of various intermediate forms of organization.

The third way this manuscript contributes to theory is by broadening the understanding of the consequences of regulatory changes, especially those implemented by social control agents, to curb misconduct (Defond and Lennox, 2017; Kedia and Rajgopal, 2011; Palmer, 2012). Specifically, our work describes how regulatory changes that reduce actors' EPRs, while discouraging misconduct in the immediate domain of

affected EPRs, may result in unintended consequences as actors respond to reductions in their opportunity set (Foss and Foss, 2022). For example, if perceived losses of EPRs shift self-employed workers from independent business to subcontracting, a host of social costs can arise, such as abuse by lead firms (i.e., the firms hiring subcontractors), subcontractors not being compensated as promised, and worker misclassification, among others (Weil, 2014). Likewise, if self-employed workers reduce their labor supply by working less or exiting the market, overall societal welfare may be reduced because buyers are forced to pay higher prices or face labor shortages. Overall, our paper's contribution is to provide an integrative framework for anticipating a wider array of responses and unintended consequences resulting from regulations that may affect actors' EPRs, thereby offering a more systematic examination than prior work (e.g., Scott et al., 2021).

6.2 Managerial and Policy Contributions

This research has numerous implications for practice. Turning first to actors in an industry that have seen an external shift occur that has weakened perceptions of EPRs, our

theory provides a means for decision-makers (whether self-employed workers or managers) to anticipate how actors subjected to reduced EPRs are likely to respond. Consider, for instance, decisions regarding firm boundaries (Tsay et al., 2018). Our theory explains why rampant shortages in 2021 and 2022 that resulted in longer lead times (Hufford, 2021) and price increases (Tita, 2021) may have contributed to some firms such as Tyson, expanding its manufacturing footprint to reduce its reliance on outside suppliers (Thomas, 2023). Within our theoretical framework, shortages reduce EPRs associated with outside procurement, and thus make vertical integration more attractive (Barzel, 1994). More broadly, our theory predicts that changes in external conditions reduce actors' perceived EPRs for products or services obtained via exchange—whether due to uncertainty about prices, quality, quantity, timely delivery, etc.—the response will be for actors to either find alternative sources or increasingly rely on vertical integration to bolster EPRs. An example of the former is how many construction firms sought out substitute materials for products with excessively long lead times during the 2021 building boom (O'Neal, 2021). An example of the latter is furniture manufacturer, wholesaler, and retailer Ashley Furniture acquiring part of the trucking fleet of Wilson Logistics in 2021 in the wake of sharp rises in the price of truckload transportation and poor service quality, as reflected in high rates of rejected shipments (Maiden, 2021; Straight, 2022). Thus, our theory points toward the role of shifting perceptions of EPRs as a key factor driving firm boundaries.²²

Another set of managerial and policy implications this research suggests is the need to consider complex interactions among multiple EPRs. Thus, it is critical to anticipate how changes in EPRs in one domain may cascade to other domains. For example, consider the offense displacement phenomena Scott et al. (2021) report whereby trucking firms that were forced to install ELDs saw an increase in unsafe driving behaviors (e.g., speeding) once enforcement of the ELD mandate began though serious HOS violations declined. Their findings are consistent with our theory in that while EPRs regarding work hours were reduced, EPRs over how trucks were driven were not affected, and thus drivers who are almost universally paid via some form of piece rate responded by driving faster (Levy, 2023). This possibility was not considered in research that examined the societal welfare implications of mandating ELDs (Pitera et al., 2013). Likewise, given many FHOOs despised the thought of being monitored electronically (Levy, 2015), it follows that the removal of EPRs due to the ELD mandate created a market for older ELD-exempt engines (Dills, 2016), despite their lower fuel economy and fifty-five-fold higher emissions as compared to modern diesel engines (Levy, 2023).

A third managerial and policy implication from our findings is that EPRs are not necessarily objective; rather, as Barzel (1994: 394) notes, EPRs take into consideration actors' perceptions of the value of the services they can obtain from

an asset directly or through market exchange. This is important because perceptions may not match reality. Consider, for instance, announcements by numerous retailers such as Target that they are closing stores in San Francisco, CA, due to rampant theft and crime (Umanzor, 2023). However, San Francisco has a much lower crime rate than Dallas, TX; Houston, TX; and Seattle, WA (amongst others) (Neilson, 2023). Thus, it is critical for policy makers to understand how actors perceive their EPRs as environmental shifts occur, especially in response to regulatory changes. This said, caution is warranted because these actors' perceptions may not mirror what ultimately transpires; for example, many dire predictions were made regarding how the ELD mandate would cause a large decrease in the number of self-employed and employee truck drivers (Levy, 2023), which did not materialize as total workers in trucking increased by 5.9% as of 2019 relative to 2016 (Bureau of Labor Statistics, 2023).

6.3 Limitations

As with all research, this manuscript has limitations. One limitation is we cannot directly observe the engine year of a truck and, instead, infer engine year based on model year. Importantly, this should reduce the power to detect effects in that some trucks in the treatment group may have ELD-exempt engines and, conversely, some trucks in the control group may have engines that require ELDs to be installed. Thus, our results cannot be attributed to this issue (and are conservative).

A second limitation, which our research shares with all quantitative studies fitting reduced-form models to archival data (Miller et al., 2023), is that we cannot directly observe the mechanisms (Astbury and Leeuw, 2010) that we postulated give rise to the empirical relations that we observed. While a limitation, it should be stressed the theory we advanced closely aligns with qualitative research from Viscelli (2018) and Levy (2023) who document how the ELD mandate reduced truckers' perceptions of the flexibility they had to utilize their vehicles. Furthermore, we can eliminate the suggestion given by Levy (2023) that inspectors were more lenient with truckers who had ELDs because we observe a spike in minor HOS violations in the postenforcement phase for treated FHOOs (see Figure 1(b)). It is unlikely that inspectors would be more stringent with minor violations but less stringent with major violations.

A third limitation relates to the quasi-experimental nature of our research design in which treatment is not randomized. Since DiD analysis guards against both differences between treatment and control groups on time-invariant covariates as well as time-varying trends homogenous across groups, any correlated omitted variables that would be able to threaten to our inferences would have to align with treatment assignment and the timing of treatment. There is a possibility that imbalance caused by nonrandom treatment assignment may raise concern that treatment and control groups might be differentially affected by such time-varying correlated omitted

variables (Atanasov and Black, 2016: 241). However, because of our deliberate choice to restrict our sample to trucks of comparable age, the disparities between treatment and control group we observe are relatively minor. For further robustness tests, we choose a sample of trucks of almost identical age to further improve covariate balance and validate that treatment and control groups are balanced with respect to preperiod trends in outcomes. Although these tests provide reassuring findings against concerns that our effect estimates are somehow biased by the presence of a time-varying correlated omitted variable, we acknowledge that we ultimately cannot rule out such a possibility.

A final limitation is that our findings are specific to a given industry. While there is strong recognition of the value of single industry studies (Joglekar et al., 2016), there are always concerns with generalizing findings from one industry to another. In this regard, we suggest those wishing to generalize our findings follow the recommendations of Miller (2023), considering whether the contextual features of a different industry are such that it is reasonable to assume the mechanisms we saw an advancement will indeed be activated to bring about similar results.

6.4 Directions for Future Research

This research can be extended in several ways. One avenue is to explore other factors that cause individuals who are self-employed and solicit their own business independently to shift to a subcontracting model. A promising candidate concerns market conditions in that industry sector, both in terms of prevailing demand as well as prices that self-employed individuals receive for their goods or services. For example, Ashe (2022) documents how surging demand and consequently rates for port drayage trucking activity in the Port of Savannah, GA, in 2021 resulted in many truck drivers leaving carriers to become FHOOs, with this process then reversing in 2022 as freight rates declined as the pace of import growth slowed. This finding is consistent with our theorizing that FHOOs will be more likely to subcontract to carriers when they place more value in carriers' marketing economies. Another route is to examine how regulatory changes in other industries that affect actors' EPRs affect organizational design choices.

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ORCID iD

Martin Holzhacker  <https://orcid.org/0000-0002-2958-1860>

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Notes

1. In additional untabulated tests, we find treated FHOOs saw a sharper decrease in inspections at fixed (and thus partially avoidable) locations versus inspections at random (and thus nearly unavoidable) locations, suggesting avoidance behaviors with the goal of restoring EPRs of operating flexibility.
2. Miller et al. (2020) use industry survey data to examine compliance efforts of small to medium-sized carriers operating between five and 100 trucks, finding that carriers with larger fleets or that operate over the wider geographic areas had higher levels of compliance in advance of the regulation's December 2017 start date. Scott et al. (2021) use large carriers known to have installed ELDs well before the mandate (and thus relatively unaffected by the regulation) as a control group for a DiD analysis of changes in HOS violations, unsafe driving violations, and crashes following mandatory ELD enforcement. They find that carriers with smaller fleets improved HOS compliance during each stage of the mandate's implementation, suggesting that the ELD mandate did have a deterrence effect. Levy (2023) provides a book-length qualitative examination of how industry stakeholders responded to the ELD mandate and highlights the limits of electronic monitoring.
3. Since ELDs are more tamper resistant than paper logs (Miller et al., 2020), any reduction in reported misconduct among affected self-employed workers is unlikely to stem from under-reporting given misconduct is more (and not less) likely to be detected when ELDs are utilized (Scott et al., 2021).
4. DOT numbers are unique carrier identification numbers assigned by the FMCSA and used to track each carrier's safety record and authorization to operate. We use census data from 2013, 2015, 2017, 2018, and 2020 to establish carrier size. If a DOT is missing from the proximate census for any inspection, we use the next most proximate census. We assume DOT numbers missing from our census data do not belong to for-hire owner-operators.
5. Regulators assume the engine year is one less than the vehicle model year. Owners of vehicles whose engine year coincides with the model year must typically file for special treatment under regulation implemented based on engine-year (see, for example, online guidance provided by the California Air Resources Board). As engine year is not observable, we exclude from our sample model-year 2000 vehicles to avoid misclassification of these vehicles.

6. Subcontracting is a binary variable equal to one for inspections under a DOT associated with a carrier that (1) has a fleet size greater than four and (2) relies on at least 80% subcontracted drivers and equal to zero otherwise.
7. Major HOS violations, minor HOS violations, and log violations are binary variables equal to one for inspections in which at least one MCMIS category 4–6, category 7, and category 2–3 violation is recorded, respectively.
8. By basing our difference-in-difference analysis on this restricted sample, we implement effectively what Atanasov and Black (2016) coin a “combined difference-in-difference/regression discontinuity” design. Atanasov and Black emphasize the synergistic nature of this combined research design:

The limited bandwidth strengthens DiD by addressing a central DiD challenge – ensuring that treated and control firms are sufficiently similar. And the DiD design strengthens RD [Regression Discontinuity Design] by controlling for any pre-shock differences between the treated and control groups. (289)

9. In an additional robustness test, we implement Mahalanobis Multivariate Distance Matching based on preperiod levels of key covariates, such as frequency of inspection, frequency of major HOS violations, frequency of major unsafe driving violations, and frequency of subcontracting. Our inferences remain unchanged for most of our hypotheses.
10. As a robustness test, we cluster standard errors by inspection state and find that our inferences remain unchanged.
11. The results are also robust to various fixed-effect—month-year, vehicle, state (untabulated), and none (untabulated)—and nonlinear logit specifications (untabulated).
12. This and all following range estimates of economic magnitude are computed using the treatment group mean preperiod levels of each respective dependent variable and the corresponding treatment effect coefficients obtained from columns 1, 2, and 3 in Tables 1–4. For example, since 3.5% of preperiod inspections result in a major HOS violation for the treatment group (see Table EC2 in the E-Companion), we estimate the economic magnitude of the treatment effect (Table 1, column 1) as $0.48/3.5 \times 100\% = 13.7\%$ percent decline in HOS violations given inspection.
13. Results presented in Table 2 are robust to various fixed-effect—month-year, truck, model year, and none—and nonlinear logit specifications (untabulated).
14. We measure inspection intensity as the number of inspections of trucks not operated by FHOOs of a given model year divided by total road-miles driven in the state in which the inspection occurs.
15. To assuage concerns that increased avoidance behaviors after the ELD mandate mentioned in footnote 1 drive reductions in inspection incidence among treated for-hire owner operators, we perform an additional untabulated robustness test in which we rerun our analysis reported in Table 2 using the subsample of unavoidable, random, roadside inspections. We continue to find significant decreases in driving frequency after the mandate effective date for owner-operators subject to the mandate ranging from 3.6% to 6.0%.
16. With the construction of the three versions of *subcontract* based on the three size terciles of carriers, the sum of the respective parameters from Columns 3, 5, and 7 (Columns 4, 6, and 8) will be similar to the parameter reported in Column 1 (Column 2), apart for slight differences due to rounding and unique estimation of the error variances and fixed effects vectors for each version of the dependent variable.
17. Coding *exit* as simply occurring in the month-year of the final inspection yields similar results.
18. On average, FHOOs are inspected every 9 months. To avoid misclassifying truck-month-years as exits in the final year of our sample (2019), we confirm that any vehicles that we classify as exiting do not appear in inspection records in the year following our sample period (2020).
19. One advantage of the Cox proportional hazard is that it does not require us to make any parametric assumptions about the baseline hazard function $h_0(m)$ (Willett, Singer, and Graham 2012).
20. The relative economic magnitude of the effect of an explanatory variable on the hazard rate in a Cox proportional-hazards model is expressed as a hazard ratio, which is approximated by exponentiating the coefficient estimate (Willett et al. 2012).
21. The results presented in Table 4 are robust to various alternative LPM specifications estimated by OLS, including fixed-effect—month-year, model year, and none (untabulated)—and nonlinear logit specifications (untabulated).
22. To further illustrate the utility of our modified PRTF, Casadesus-Masanell and Spulber (2000) explain that the main reason General Motors acquired Fisher Body was to better ensure a supply of closed metal bodies given strong demand growth for cars with closed bodies (see their Table 2 and Table 3). Within our theory, the growing importance of closed bodies can be thought as weakening GM’s perceived EPRs when engaging with Fisher Body as an outside supplier because GM feared that supply would not be available if its rivals also began purchasing from Fisher Body. Thus, our theory can account for this historical example, whereas canonical PRTF cannot as Casadesus-Masanell and Spulber (2000: 75–76) describe in their critique of Hart’s (1995) evaluation of this case.

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